Course Title: Cryptography and Network Security Lab	**Course Code**: BCSE309P
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CEASAR CYPHER (SHIFT CYPHER)

Code

Sender

```
// client

import java.net.*;
import java.io.*;
import java.util.*;

public class sender {
    public static void main(String[] args) throws Exception {

        Scanner a = new Scanner(System.in);

        System.out.println("Client started...");
        System.out.println("Connecting to server...");
        Socket socket = new Socket("localhost", 5000);
        System.out.println("Connected to server.");
        System.out.print("Enter the string: ");
        String pt = a.nextLine();
        System.out.print("Enter the key: ");
        int k = a.nextInt();
        String ct = "";

        for (int i = 0; i < pt.length(); i++) {
            char c = pt.charAt(i);
            if (Character.isLetter(c)) {
                int shift = k % 26;
                if (Character.isLowerCase(c)) {
                    ct+= (char) ((c - 'a' + shift) % 26 + 'a');
                } else {
                    ct += (char) ((c - 'A' + shift) % 26 + 'A');
                }
            } else {
                ct += c;
            }
        }
```

```
        System.out.println("Cipher text: " + ct);

        PrintWriter out = new PrintWriter(socket.getOutputStream(), true);

        System.out.println("Sending encrypted message and key to server...");
        out.println(ct);
        out.println(k);

        socket.close();
        a.close();
    }
}
```

Receiver

```
// server

import java.net.*;
import java.io.*;

public class receiver {
    public static void main(String[] args) throws Exception {

        System.out.println("Server starting...");
        ServerSocket serverSocket = new ServerSocket(5000);
        System.out.println("Server waiting for client connection...");

        Socket socket = serverSocket.accept();
        System.out.println("Client connected.");

        BufferedReader in = new BufferedReader(
                new InputStreamReader(socket.getInputStream()));

        String ct = in.readLine();
        int k = Integer.parseInt(in.readLine());

        System.out.println("Cipher text received: " + ct);
        System.out.println("Key received: " + k);

        String pt = "";

        for (int i = 0; i < ct.length(); i++) {
            char c = ct.charAt(i);

            if (Character.isLetter(c)) {
                int shift = k % 26;
```

```
                if (Character.isLowerCase(c)) {
                    pt += (char) ((c - 'a' - shift + 26) % 26 + 'a');
                } else {
                    pt += (char) ((c - 'A' - shift + 26) % 26 + 'A');
                }
            } else {
                pt += c;
            }
        }

        System.out.println("plain text: " + pt);

        socket.close();
        serverSocket.close();
    }
}
```

Output [screenshot]

Sender

```
PS E:\aa vit sem6> javac sender.java
PS E:\aa vit sem6> java sender
Client started...
Connecting to server...
Connected to server.
Enter the string: meet me tomorrow
Enter the key: 3
Cipher text: phhw ph wrpruurz
Sending encrypted message and key to server...
PS E:\aa vit sem6>
```

Receiver

```
PS E:\aa vit sem6> javac receiver.java
PS E:\aa vit sem6> java receiver
Server starting...
Server waiting for client connection...
Client connected.
Cipher text received: phhw ph wrpruurz
Key received: 3
plain text: meet me tomorrow
PS E:\aa vit sem6>
```

VIGENERE CIPHER

Code

Sender

```java
// client

import java.net.*;
import java.io.*;
import java.util.*;

public class sender_vignere {
    public static void main(String[] args) throws Exception {

        Scanner a = new Scanner(System.in);

        System.out.println("client started...");
        System.out.println("connecting to server...");

        Socket s = new Socket("localhost", 5000);
        System.out.println("Connected to server.");

        System.out.print("Enter the string: ");
        String pt = a.nextLine();

        System.out.print("Enter the key: ");
        String k = a.nextLine();

        String ct = "";
        int keyIndex = 0;

        for (int i = 0; i < pt.length(); i++) {
            char m = pt.charAt(i);

            if (Character.isLetter(m)) {
                char c = k.charAt(keyIndex % k.length());
                int shift = Character.toLowerCase(c) - 'a';

                if (Character.isLowerCase(m)) {
                    ct += (char) ((m - 'a' + shift) % 26 + 'a');
                } else {
                    ct += (char) ((m - 'A' + shift) % 26 + 'A');
                }
                keyIndex++;
            } else {
```

```
                ct += m;
            }
        }

        System.out.println("cipher text: " + ct);

        PrintWriter out = new PrintWriter(s.getOutputStream(), true);

        System.out.println("Sending cipher text and key to server...");
        out.println(ct);
        out.println(k);

        System.out.println("Message sent. Closing connection.");

        s.close();
        a.close();
    }
}
```

Receiver

```
// server

import java.net.*;
import java.io.*;

public class receiver_vignere {
    public static void main(String[] args) throws Exception {

        System.out.println("Server starting...");
        ServerSocket ss = new ServerSocket(5000);
        System.out.println("Server waiting for client connection...");

        Socket s = ss.accept();
        System.out.println("Client connected.");

        BufferedReader in = new BufferedReader(
                new InputStreamReader(s.getInputStream()));

        String ct = in.readLine();
        String k = in.readLine();

        System.out.println("cipher text received: " + ct);
        System.out.println("Key received: " + k);

        String pt = "";
        int keyIndex = 0;
```

```java
        for (int i = 0; i < ct.length(); i++) {
            char c = ct.charAt(i);

            if (Character.isLetter(c)) {
                char ch = k.charAt(keyIndex % k.length());
                int shift = Character.toLowerCase(ch) - 'a';

                if (Character.isLowerCase(c)) {
                    pt += (char) ((c - 'a' - shift + 26) % 26 + 'a');
                } else {
                    pt += (char) ((c - 'A' - shift + 26) % 26 + 'A');
                }
                keyIndex++;
            } else {
                pt+= c;
            }
        }

        System.out.println("plain text: " + pt);
        System.out.println("closing connection.");

        s.close();
        ss.close();
    }
}
```

Output [screenshot]

Sender

```
PS E:\aa vit sem6> javac sender_vignere.java
PS E:\aa vit sem6> java sender_vignere
client started...
connecting to server...
Connected to server.
Enter the string: wearediscoveredsaveyourself
Enter the key: deceptive
cipher text: zicvtwqngrzgvtwavzhcqyglmgj
Sending cipher text and key to server...
Message sent. Closing connection.
PS E:\aa vit sem6>
```

Receiver

```
PS E:\aa vit sem6> javac receiver_vignere.java
PS E:\aa vit sem6> java receiver_vignere
Server starting...
Server waiting for client connection...
Client connected.
cipher text received: zicvtwqngrzgvtwavzhcqyglmgj
Key received: deceptive
plain text: wearediscoveredsaveyourself
closing connection.
PS E:\aa vit sem6>
```

VERNAM CYPHER

Code

Sender

```
// client

import java.net.*;
import java.io.*;
import java.util.*;

public class sender_vernem {
    public static void main(String[] args) throws Exception {

        Scanner a = new Scanner(System.in);

        System.out.print("Enter the plaintext: ");
        String pt = a.nextLine();

        System.out.print("Enter the key (same length): ");
        String k = a.nextLine();

        if (pt.length() != k.length()) {
            System.out.println("Not same length");
            a.close();
            return;
        }

        System.out.println("Client started...");
        System.out.println("Connecting to server...");

        Socket s = new Socket("localhost", 5000);
        System.out.println("Connected to server.");

        String ct = "";

        for (int i = 0; i < pt.length(); i++) {
            char m = pt.charAt(i);
            char c = k.charAt(i);

            if (m >= 'a' && m <= 'z') {
                ct += (char) (((m - 'a') ^ (c - 'a')) + 'a');
            } else {
                ct += m;
            }
        }

        System.out.println("Cipher Text (CT): " + ct);
```

```
        PrintWriter out = new PrintWriter(s.getOutputStream(), true);

        out.println(ct);
        out.println(k);

        System.out.println("Message sent. Closing connection.");

        s.close();
        a.close();
    }
}
```

Receiver

```
// server

import java.net.*;
import java.io.*;

public class receiver_vernem{
    public static void main(String[] args) throws Exception {

        System.out.println("Server starting...");
        ServerSocket ss = new ServerSocket(5000);
        System.out.println("Server waiting for client connection...");

        Socket s = ss.accept();
        System.out.println("Client connected.");

        BufferedReader in = new BufferedReader(
                new InputStreamReader(s.getInputStream()));

        String ct = in.readLine();
        String k = in.readLine();

        System.out.println("Cipher Text received: " + ct);
        System.out.println("Key received: " + k);

        String pt = "";

        for (int i = 0; i < ct.length(); i++) {
            char m = ct.charAt(i);
            char c = k.charAt(i);

            if (m >= 'a' && m <= 'z') {
                pt += (char) (((m - 'a') ^ (c - 'a')) + 'a');
```

```java
            } else {
                pt += m;
            }
        }

        System.out.println("Plain Text: " + pt);
        System.out.println("Closing connection.");

        s.close();
        ss.close();
    }
}
```

Output [screenshot]

Sender

```
PS E:\aa vit sem6> javac sender_vernem.java
PS E:\aa vit sem6> java sender_vernem
Enter the plaintext: hello
Enter the key (same length): money
Client started...
Connecting to server...
Connected to server.
Cipher Text (CT): lkgpw
Message sent. Closing connection.
PS E:\aa vit sem6>
```

Receiver

```
PS E:\aa vit sem6> javac receiver_vernem.java
PS E:\aa vit sem6> java receiver_vernem
Server starting...
Server waiting for client connection...
Client connected.
Cipher Text received: lkgpw
Key received: money
Plain Text: hello
Closing connection.
PS E:\aa vit sem6>
```